B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:	II						
Paper Category:	Practical **(Major) (Group-I)**						
Paper Name:	Practical based on DSC3-2 **Paper Code :** G06-0203-P						
Credit:	02	**Practical:**	4 Hrs./Week				
Marks:	**UA:**	30	**CA:**	20	**Total:**	50	

Software: Keil uVision IDE, Proteus for simulation, Any text editor

1. Create a simple project in Keil MicroVision. Write, compile, and debug a basic program to control an LED. Simulate the program and analyze the results.
2. Download and install Flash Magic from the official website. Launch the Flash Magic application. Select the target microcontroller. Configure the communication settings (COM port, baud rate, etc.). Load the compiled HEX file generated from Keil MicroVision.
3. Write a program to demonstrate the use of internal memory, registers, and Special Function Registers (SFRs) in the 8051 microcontrollers.
4. Write and execute ALP to add, subtract, two 8 bit numbers.
5. Write and execute ALP to multiply, and divide two 8 bit numbers.
6. Write and execute ALP to implement logicaloperations like AND, OR, XOR on data.
7. Write a program to configure and use the ports of the 8051 microcontrollers.
8. Write a program to configure and use the timer/counter functionality of the 8051microcontrollers.
9. Write and test simple programs to use timers for delay generation.
10. Write a program to configure and use the serial port of the 8051 microcontrollers forserial communication.
11. Write simple programs to demonstrate data transfer instructionsusing the 8051-instruction set .
12. Write simple programs to demonstrate arithmetic, and logical instructions using the 8051-instruction set
13. Write a program to turn an LED on and off using a port pin of the 8051 microcontrollers.
14. Write a program to read input from one port and send output to another port using

the8051 microcontrollers.

15. Write a program to blink an LED at regular intervals using the timer functionality ofthe 8051 microcontrollers.
16. Write small assembly programs using different types of instructions.
17. Write a program to send data from the 8051 to a PC via the serial port.
18. Write a program to display numbers 0-9 on a 7-segment display.
19. Implement a programto create a simple counter using the 7-segment display.
20. Write a program to display characters and strings on an LCD. Create a program to scrolltext on the LCD.
21. Write a program to read temperature data from the sensor. Display the temperature onan LCD.
22. Write a program to rotate the stepper motor in different directions. Implement speedcontrol for the stepper motor.